

The Effect of Gunshot Detection Technologies on Policing Practices: An Empirical Examination of the Chicago Police Department, Burtch, Greenwood & Watson

I chose this article because I believe it is the closest to the project I imagined we will be working on, and reviewing it in detail would help me with designing the Flock Camera Project.

What is the treatment, and what is the treated/control unit?

- My understanding is that the treatment is “ShotSpotter becoming operational in a police district”
 - This means that a treated unit is a police district with ShotSpotter operational in its area
 - The control units are districts that have not yet received ShotSpotter.
 - Overall, the unit of analysis is a police district-month, because researchers observe each district monthly.

What creates the comparison between treated and control units?

- The researchers compare changes in different concepts using different operational measures in order to study the changes and differences between the treatment group and control group
 - The comparison comes from ShotSpotter becoming operational in different districts at different times.
 - Researchers compare changes in policing outcomes before and after deployment with changes over the same period in districts that had not yet received ShotSpotter

Where does the identification come from?

- I believe the identification comes from differences in deployment timing. Since identification means figuring out whether the treatment actually caused the outcome, the researchers compare changes in districts that received ShotSpotter with changes in districts that had not yet received it or did not receive it during the study.

What is the key identifying assumption?

- I think the key assumption is that without ShotSpotter implementation, policing outcomes in treated districts would have changed similarly to those in control districts.

What would violate that assumption or otherwise bias the estimate?

- I think there are a couple of different ways that this assumption could fail or bias the estimate
 - If the treated districts and the control districts were already following different trends, or, in other words, are not comparable to begin with
 - If another change, such as a new policing policy, happened there around the same time as ShotSpotter deployment
 - Another issue is that the treatment is causing something else in parallel; for example, it could shift crime between the two groups.

What does the estimated coefficient actually represent?

- The estimated coefficient represents the average effect of ShotSpotter on districts that received it (because receiving and not receiving is 0/1)
 - For example, the stop estimate suggests that ShotSpotter reduced recorded police stops by about 17.6%, assuming the study's identifying assumptions hold.

What are the main limitations of the design?

- We can say that quasi-experimental designs have limitations, such as the fact that districts that received the deployment were not totally random.
- There are also some limitations with the measurements for each concept
 - For example, recorded stops miss unrecorded encounters.
- In terms of generalizability, results from Chicago may not apply to cities with different policing practices.
- Also, there seems to be a missing link between an individual alert and its related searches or stops. (I'm not sure if it matters in the big picture, but it might help explain exactly why behavior changed.)

What solutions or methodological approaches are used to address these biases or limitations, and why are they needed?

- There can be stricter measures to make sure the research accounts for stable differences between districts and changes affecting the whole city.
 - Make sure the outcomes were already changing differently before deployment
- It would be beneficial to look deeper into the treatment group and how districts in that group are different
 - Check whether one unusual district drives the results
- In addition to the DiD they did for this research, it would be useful to compare normal patterns using data from districts before they receive ShotSpotter and after. This is not by itself methodologically sufficient but can help as an additional stream of comparison.

How does the setup change when treatment occurs at different times across units?

- When treatment starts at different times across units, the setup becomes a staggered difference-in-differences
 - Each district has its own before-and-after timeline. We compare districts during the same years, using districts that have not received the technology yet as controls.

Based on what you learn, how might the design apply to our Flock camera setting?

- Firstly, I think the general methodology of DiD and how it is conducted would be similar to the Flock Camera setting.
 - The pivotal point of their DiD is the deployment date, and in our case we only have estimates of this using different data sources, and it's not concrete.
- The way they developed the panel data is something I'm hoping to be able to do with our data. It is a little more complicated in our case because districts share their data with each other as well.
- Also, I think the measurements they developed could be used in our project as well. We might need to develop new measurements that are more descriptive of the privacy/ethical concerns with the Flock Cameras.
- Our DiD might also be staggered because of the adoption dates.

